Ram Prasad Bidiyasar

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Backend Software Engineer
Mobile: 8209641119
Java | Spring Boot | Distributed Systems | Kafka | ScyllaDB | Redis | Aerospike | GCP

Summary

Backend Software Engineer with nearly 3 years of experience building distributed systems, high-throughput microservices, and low-latency backend platforms using Java, Spring Boot, Kafka, and WebFlux. Owns production systems serving millions of users across backend architecture, deployment, observability, incident response, and performance optimization. Contributed to production-scale AWS to GCP cloud migration for 20+ Airtel Xstream microservices, including Kafka/SQS, Kinesis, S3, MongoDB, ScyllaDB, and related workloads while maintaining production stability.

Technical Skills

Languages: Java 8/11/17/21, SQL
Backend: Spring Boot, Spring WebFlux, Spring Cloud, REST APIs, gRPC, Reactive Programming
Distributed Systems: Kafka, Event-Driven Architecture, Microservices, System Design, Scalability, High Availability
Databases: ScyllaDB, Aerospike, Redis, MongoDB, PostgreSQL
Cloud: GCP, GCS, AWS, Cloud Migration
Observability: Metrics, Prometheus, Grafana, Production Reliability, Performance Optimization
DevOps: Docker, CI/CD, Fault Tolerance, Circuit Breaker, Retry, Idempotency, Rate Limiting, Load Balancing

Experience

Senior Software Engineer June 2025 – Present
Airtel *Gurugram,India*

- Owned backend architecture and production reliability for Airtel Xstream batch, content, play, sync, and user microservices powering playback, ingestion, user journeys, and partner integrations across Java, Spring Boot, Kafka, Redis, Aerospike, MongoDB, and ScyllaDB.
- Optimized personalized page generation for Xstream web/mobile clients using Java 21, Spring Boot, gRPC, thread-pool tuning, concurrency, and parallel asynchronous calls, scaling throughput from 30 to 100 TPS and reducing latency by nearly 30%.
- Designed a distributed IPTV session platform for single-device streaming with opaque session tokens and reusable SDK-based validation across SMS and Playback services, improving fault tolerance on Java 21, Spring Boot, and Aerospike.
- Re-architected watch-history processing with Spring WebFlux and ScyllaDB reactive programming, reducing P99 latency by 75% and improving per-instance throughput by approximately 4× for high-scale user activity workloads.
- Engineered Multi-CDN routing and failover for IPTV playback used by 10M+ active users, evaluating MongoDB-configured Java/SpEL rules across geo, user, channel, priority, weights, and CDN health to reduce buffering from 1% to 0.07%.
- Led production ownership for the UCLM campaign platform across 20+ Kafka/Spring Boot microservices processing 1.5B+ daily messages, driving observability, capacity planning, deployments, incident response, scalability, and active development with OracleDB, Airflow, Flink, and Druid.

Software Engineer Oct. 2023 – June 2025
Airtel *Gurugram, India*

- Redesigned Continue Watching synchronization with a ScyllaDB three-store model, table-level compaction, app-level LRU eviction, Redis/Aerospike distributed caching, bucket-wise diff-sync merge, Kafka asynchronous persistence, and local fallback, reducing sync latency by nearly 70%.
- Implemented event-driven episode promotion using Kafka and Java/Spring Boot through `syncEpisodicLatestContent`, surfacing newly ingested episodes for watched OTT series from summaries while avoiding full ScyllaDB history scans.
- Delivered production partner integrations across Xstream microservices, onboarding Amazon Prime playback via App-in-App deep links, subscription/claim/login flows, and scheduled Hotstar TV, movie, VOD, and live ingestion through Spring Boot REST APIs and batch-service.
- Contributed to AWS to GCP migration for 20+ Airtel Xstream production microservices, replacing SQS/Kinesis with Kafka, S3 with GCS, and migrating MongoDB and ScyllaDB workloads while coordinating rollout, minimizing downtime, and preserving production continuity.

Projects

Aerospike Cluster Management Platform | *Java 21, Spring Boot 3.2, Aerospike, React 19, Vite*

- Implemented a production-grade Aerospike cluster operations platform with Spring Boot services and React/Vite UI, supporting record CRUD, data browsing, health monitoring, and diagnostics.

Reactive API Gateway Platform | *Java 21, Spring Cloud Gateway, MongoDB, JWT, Docker*

- Designed a production-grade reactive API gateway with MongoDB-backed dynamic routing, JWT authentication/authorization, custom filters, correlation IDs, admin APIs, and Docker deployment.

Education

Malaviya National Institute of Technology Jaipur, Rajasthan
B.Tech. in Electronics and Communication Engineering | CGPA: 8.16 *July 2019 – May 2023*